

# Performance Testing

|              |                                          |               |                          |
|--------------|------------------------------------------|---------------|--------------------------|
| Date         | 30 September 2026                        | Team ID       | 6ab3e57da238dc999eb3d44d |
| Project Name | EduGenie – AI-Powered Learning Assistant | Maximum Marks | As per template          |

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**GitHub:** <https://github.com/ricardovictor-28507/Google-Cloud-Generative-AI--A-S>

Important note: the current project was verified with a local smoke test, but no dedicated load-testing tool was run in the available project evidence. The performance table therefore distinguishes targets from measured values rather than inventing benchmark results.

## Step 1: Testing Overview

| Field               | Details                                                                |
|---------------------|------------------------------------------------------------------------|
| Testing Tool Used   | Local Uvicorn + curl smoke tests; dedicated load test not yet executed |
| Type                | Functional smoke testing; performance benchmark planned                |
| Target Module / API | `/health`, `/explain` and all five workflow endpoints                  |
| Environment         | Local development environment                                          |
| Test Date           | 30 September 2026                                                      |

## Step 2: Test Scenarios

| S.No | Scenario                       | Users | Duration       | Expected Outcome              |
|------|--------------------------------|-------|----------------|-------------------------------|
| 1    | GET /health                    | 1     | Single request | HTTP 200 and status ok        |
| 2    | POST /explain with valid topic | 1     | Single request | HTTP 200 with explanation     |
| 3    | POST /quiz with valid text     | 1     | Single request | Structured quiz list          |
| 4    | Empty / oversized request      | 1     | Single request | HTTP 422 validation response  |
| 5    | AI unavailable / no key        | 1     | Single request | Demo fallback or useful error |

## Step 3: Performance Results

| Metric                | Target            | Actual       | Status       | Remarks                                              |
|-----------------------|-------------------|--------------|--------------|------------------------------------------------------|
| Average response time | To be benchmarked | Not measured | Not measured | Depends on local runtime and Gemini network latency. |
| Maximum response time | To be benchmarked | Not measured | Not measured | Run JMeter/Locust/k6 before production deployment.   |
| Throughput            | To be benchmarked | Not measured | Not measured | No concurrency test evidence included.               |
| Error rate            | < 1% target       | Not measured | Not measured | Functional error paths were checked locally.         |
| CPU / memory          | < 80% target      | Not measured | Not measured | Production environment not defined.                  |

## Step 4: Observations

The local application starts successfully, `/health` returns a valid response, and the Explain workflow returns a demo response without a live Gemini key. Live-AI performance must be measured separately because external model latency is variable.